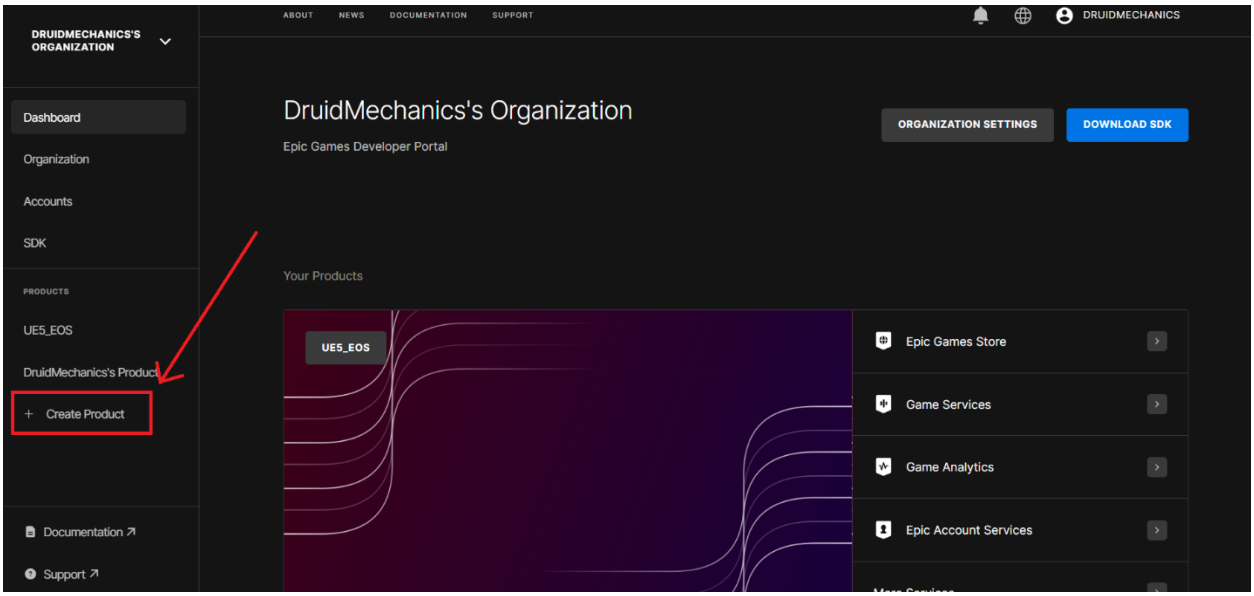
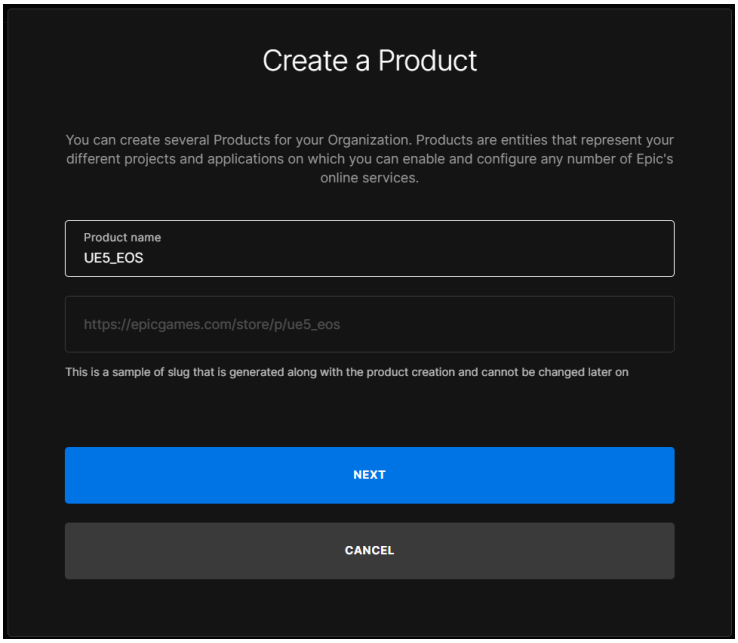
1. EOS Product

To use EOS (Epic Online Services) for your project, you must create an EOS product.

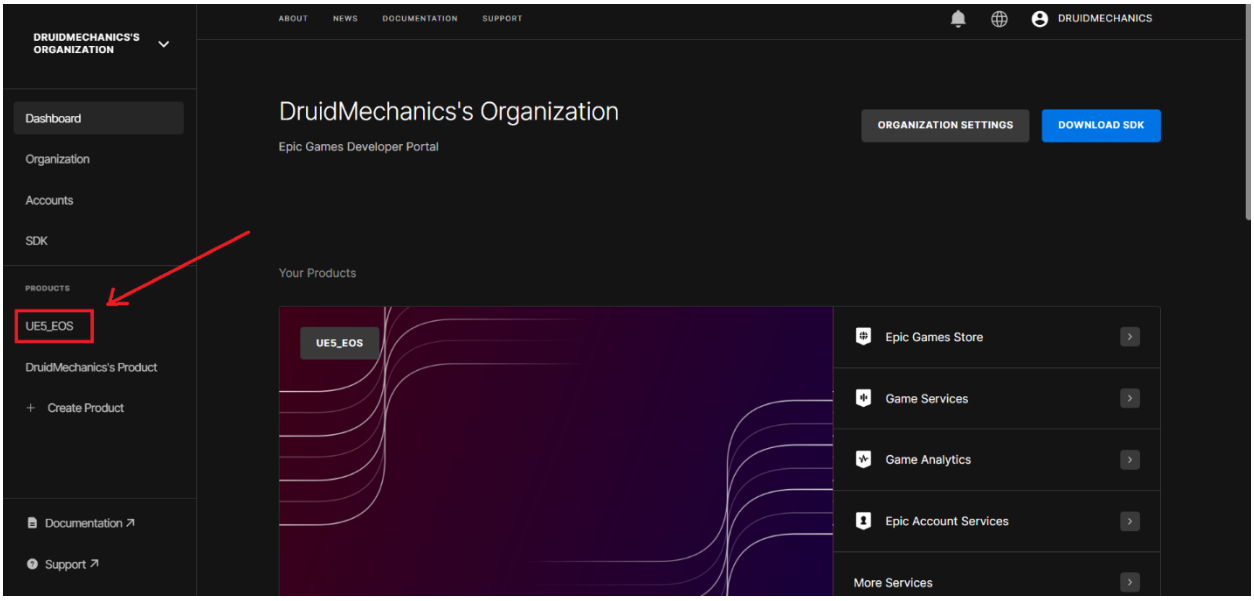
- Go to dev.epicgames.com
- Log into your Epic Games account
- Click on Create Product



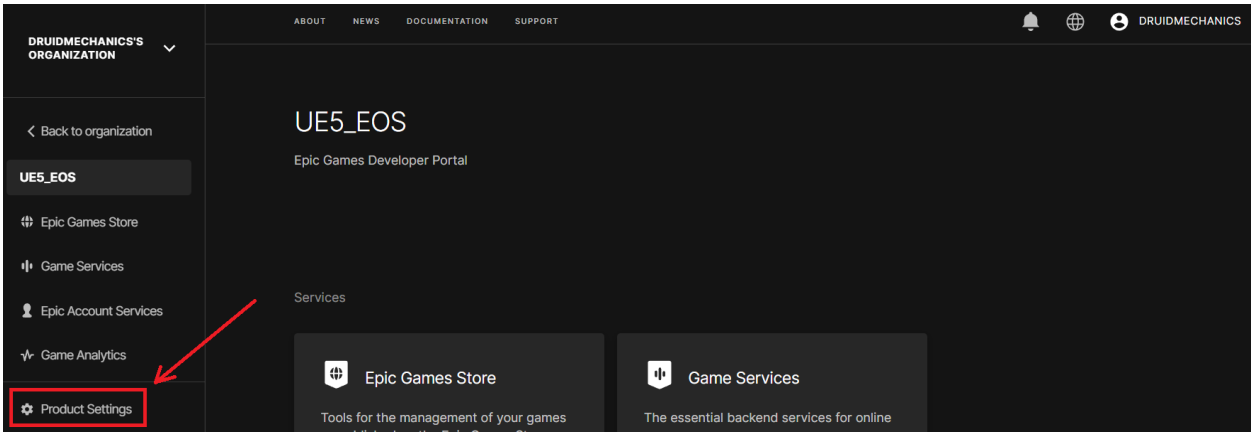
- Enter a Product Name (this can't be changed later) and click Next



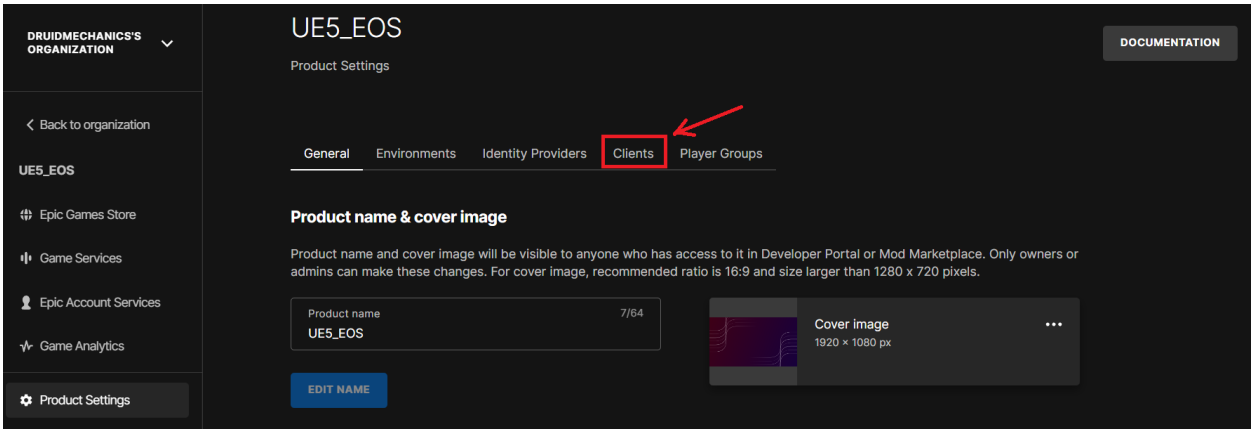
- (optional) upload a cover image
- Click on your new product



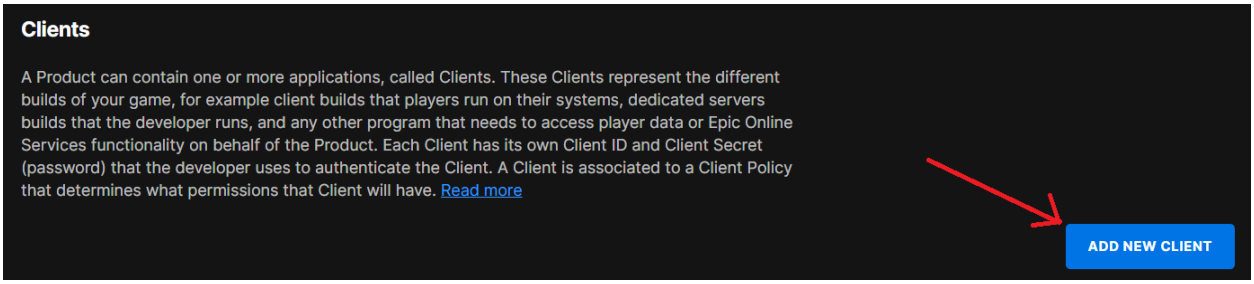
- Go to Product Settings



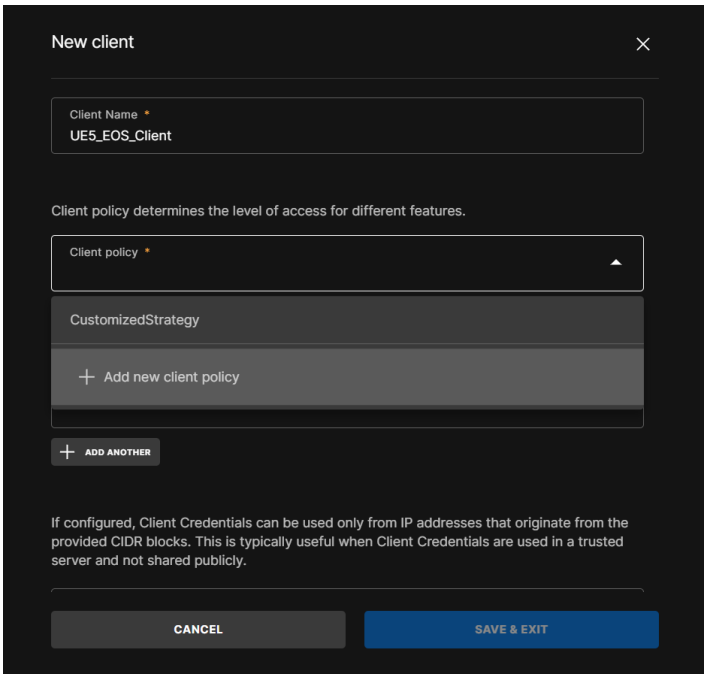
- Click the Clients tab



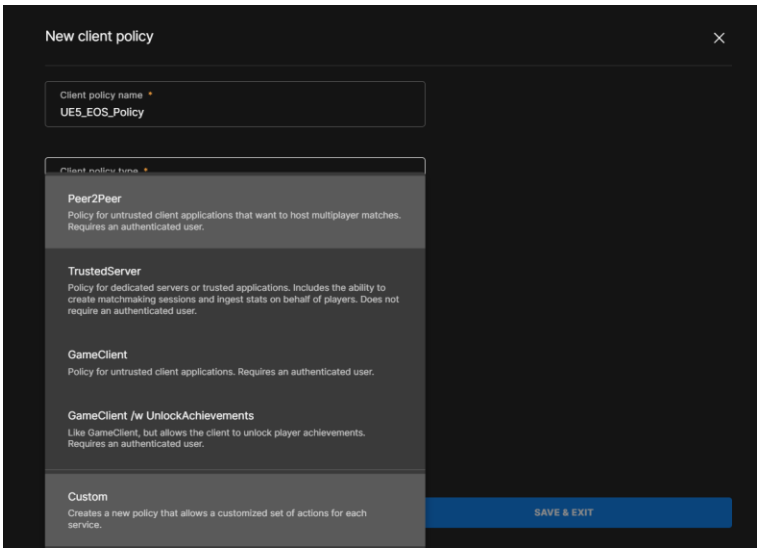
- Click Add New Client



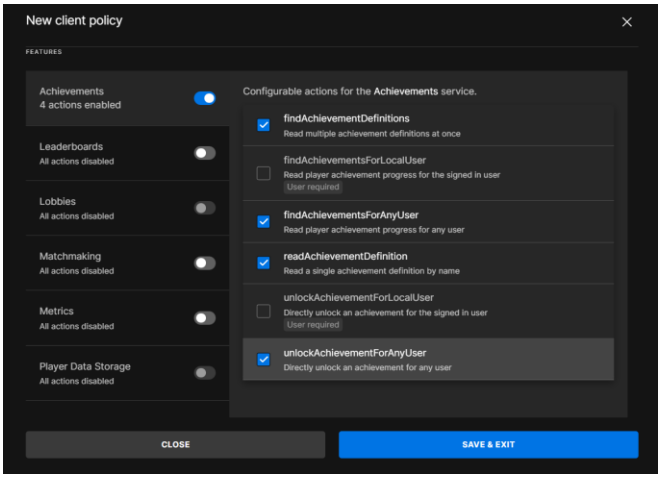
- Enter a Client name, then open the drop-down for Client Policy and choose + Add New Client Policy



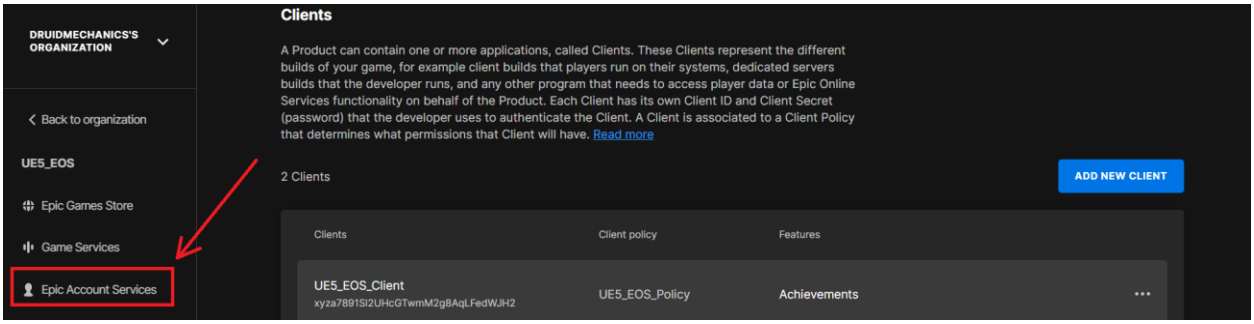
- Give your policy a name and choose a policy type. You can choose Custom



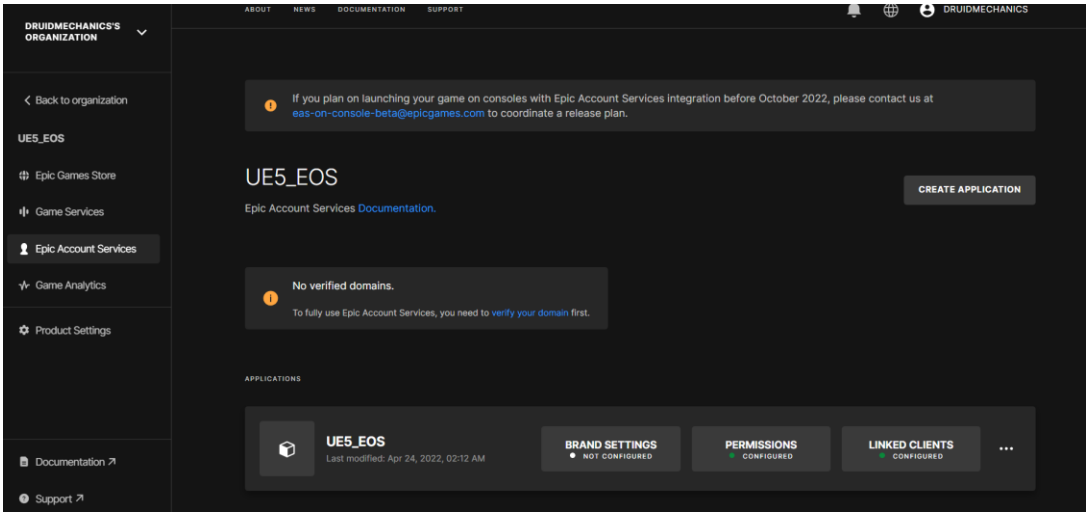
- Enable all features you would like to have in your product
    - Each feature has a list of sub-features. Check the checkboxes to enable those
- Features that say User Required means an authenticated user must be logged in to access those features. You can leave those uncheckd.



- When finished, click Save & Exit from the policy creation
- Click Save & Exit from the New Client Creation
- Next, click Epic Account Services

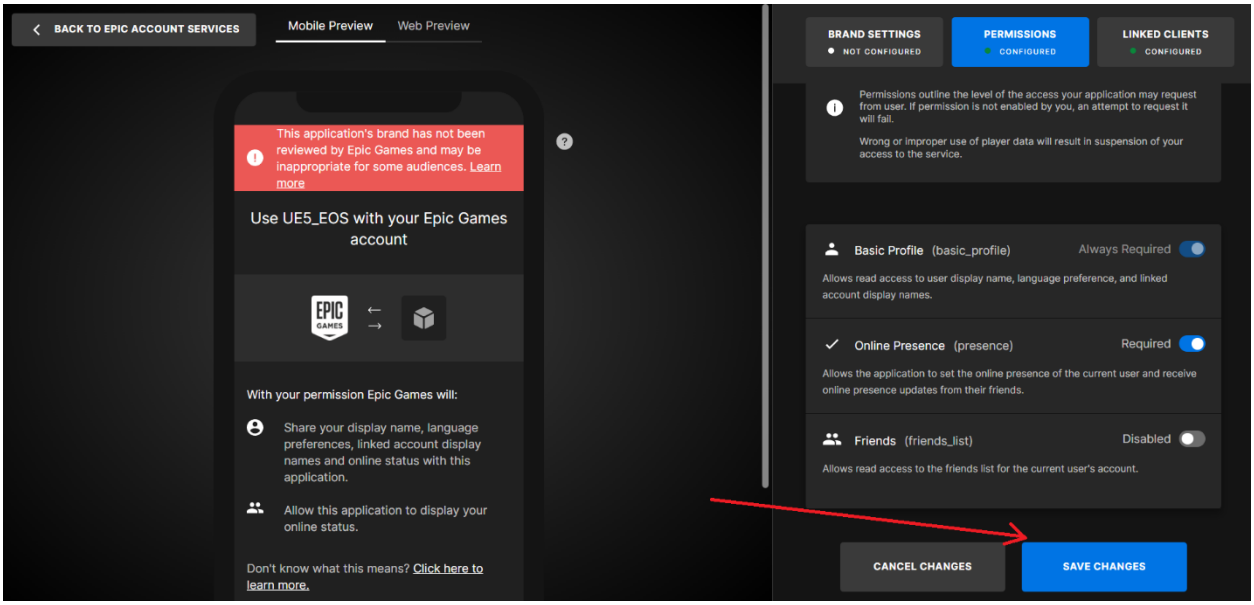


There are three parts to configuring an Epic Account Service: Brand Settings, Permissions, and Linked Clients

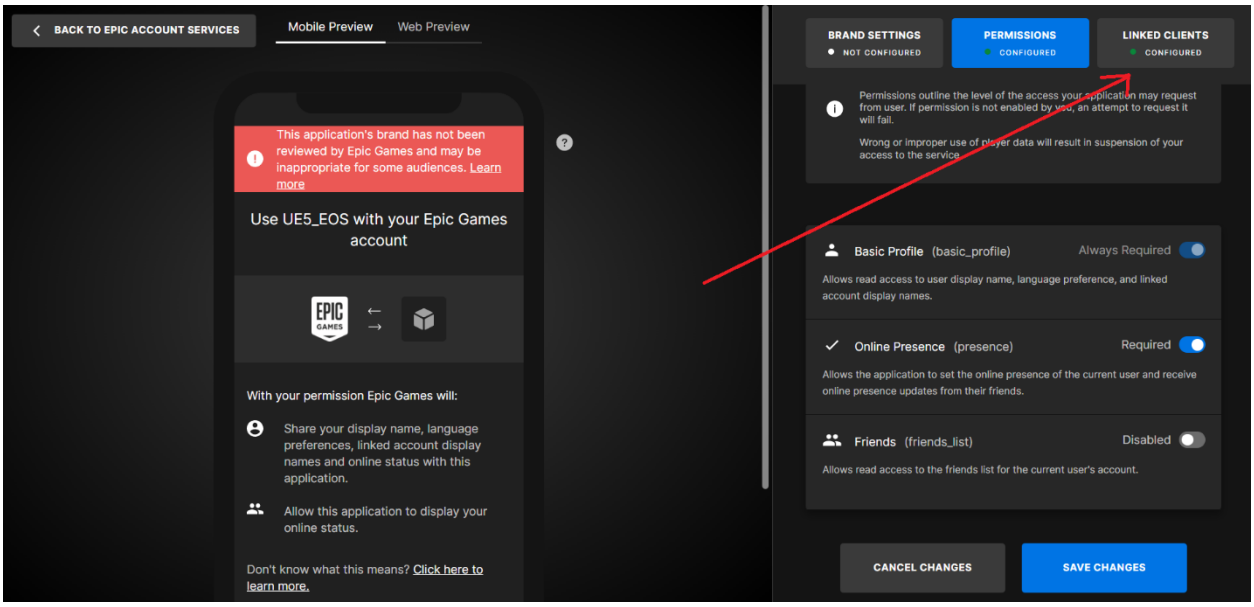


Permissions

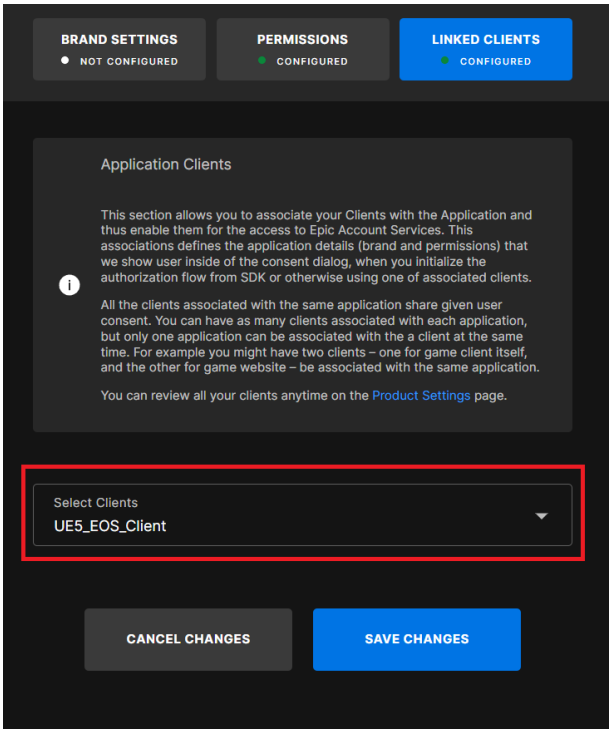
- Enable the permissions you need and Save Changes



- Select Linked Clients



- Choose the Client you created earlier



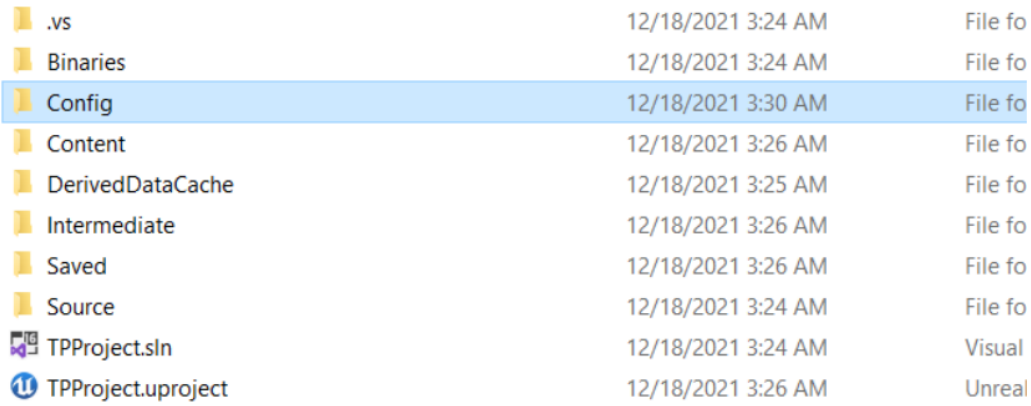
- Click Save Changes
- Brand Settings can be skipped for now, as this is a demonstration. When ready, you can submit your Brand settings to Epic for Review.

2. Create or open an Unreal Engine 5 Project

(For new projects, it is recommended that you use the Third Person project template, as it already has a working character which is replicated).

3. Configure the project for Epic Online Services (EOS)

- Go to Edit -> Plugins -> search for Online Subsystem EOS and EOS Shared
- Check Enabled for both
- Click Restart Now
- In the project folder, go to Config



- Open DefaultEngine.ini

Add the following to the bottom of DefaultEngine.ini (copy them directly from Unreal Engine's Documentation page) https://docs.unrealengine.com/4.27/en-US/ProgrammingAndScripting/Online/EOS/

```
[OnlineSubsystemEOS]
bEnabled=true

[OnlineSubsystem]
DefaultPlatformService=EOS

[/Script/Engine.GameEngine]
+NetDriverDefinitions=(DefName="GameNetDriver",DriverClassName="OnlineSubsystemEOS.NetDriverEOS",DriverClassNameFallback="OnlineSubsystemUtils.IpNetDriver")

[/Script/OnlineSubsystemEOS.NetDriverEOS]
bIsUsingP2PSockets=true
```

EOS Project Settings

In Project Settings -> Plugins -> Online Subsystem EOS, check:

EOS Plus Login Settings

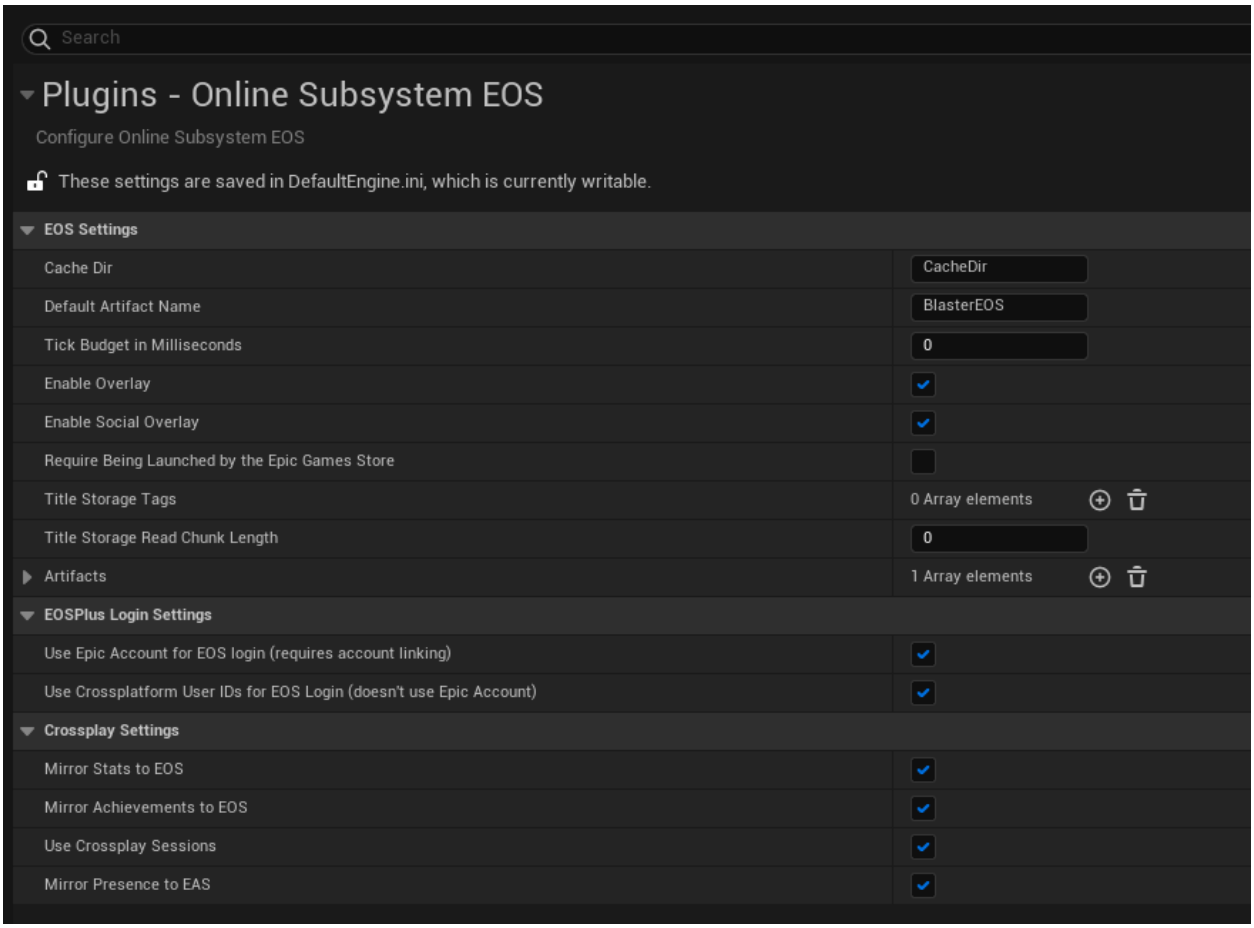
- Use Epic Account for EOS login
- Use Crossplatform User IDs for EOS Login

Crossplay Settings

- Mirror Stats to EOS
- Mirror Achievements to EOS
- Use Crossplay Sessions
- Mirror Presence to EAS

EOS Settings

- Enable Overlay
- Enable Social Overlay

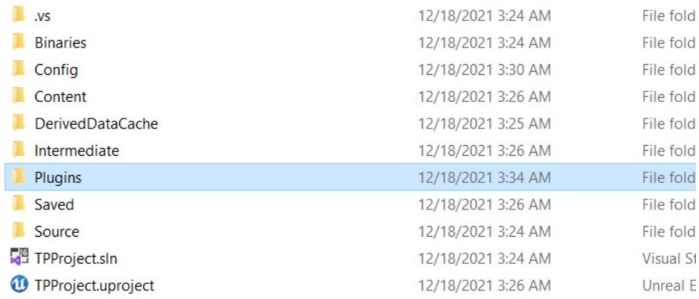


(**not** Require Being Launched by the Epic Games Store)

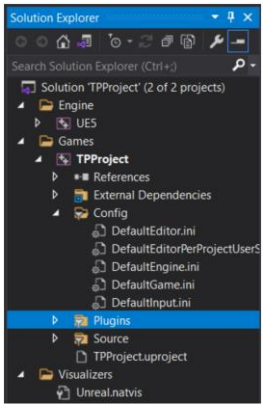
- Add an element to the Artifacts array. Open the dropdown and fill it out.
- In EOS Settings, set a Default Artifcat Name: BlasterEOS
- In the element in Artifacts, set the Artifact Name: BlasterEOS
- Fill in all your IDs; you can ignore encryption key.

4. Add the Multiplayer Plugin

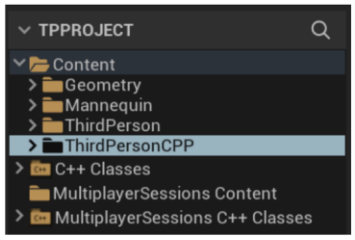
- Copy the Plugins folder into your project:



- Inside Plugins, you should see MultiplayerSessions
- Close Unreal Engine and Visual Studio. Delete your Binaries, Intermediate, and Saved folders. Right-click your .uproject and select Generate Visual Studio project files.
- Open your Visual Studio project solution (.sln). You should now see a Plugins folder in the Solution Explorer



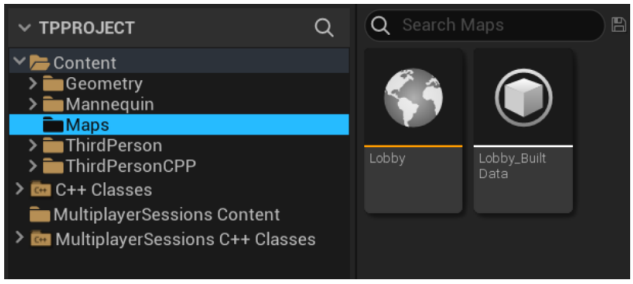
- Open your project in Unreal Engine. It will ask if you would like to rebuild modules. Select Yes.
- In the Content Browser, click Settings. Check Show Plugin Content.
- You will now have two extra folders in the Content Browser: Multiplayer Sessions Content and Multiplayer Sessions C++ Classes.



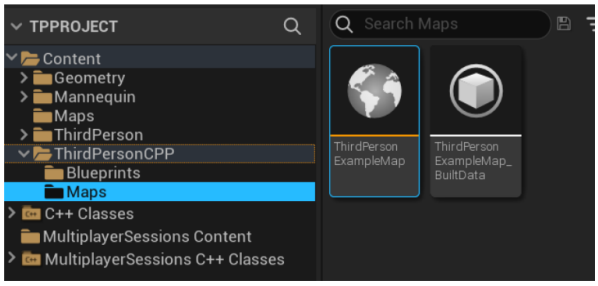
- Go to the Multiplayer Sessions Content folder. There is a Widget Blueprint called WBP_Menu.

Create a Lobby Level

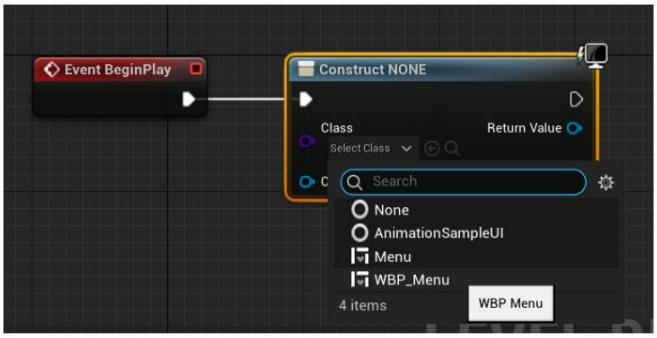
- Create a new level, and save it in your project (it doesn't matter where you save it or what you name it)



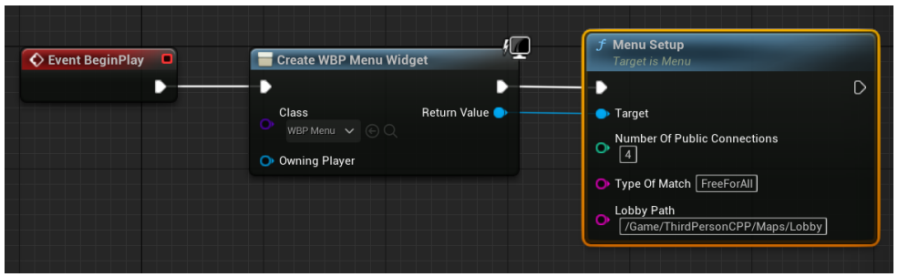
- Open your starting level



- Open the Level Blueprint. Use Create Widget to create a new widget, selecting WBP_Menu as the widget class. Connect this to Begin Play.



- Drag off of the Return Value output and search for Menu Setup. Select it.

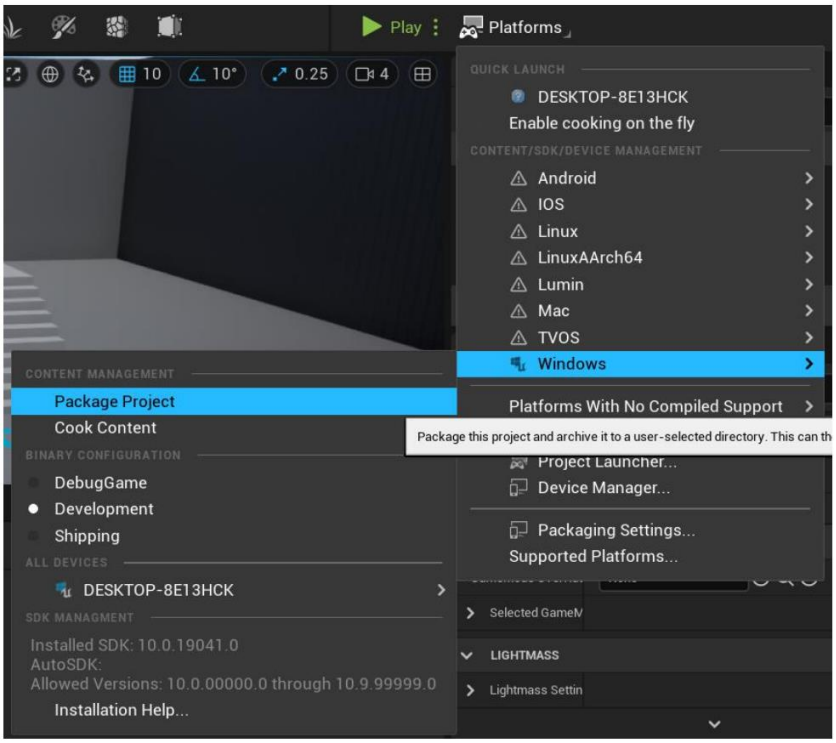


- Change the Number of Public Connections to the number of players you want in your game
- Change the lobby path to the path to your lobby level (use /Game/ instead of /Content/) and leave out the .umap extension

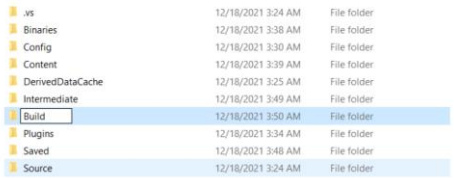


Play Test the Game

- Go to Platforms -> -> Package Project.



- Create a new folder or select a folder destination





- Click Select Folder
- Upload to a Google drive (or anywhere else) and download it onto a second machine
- Launch one machine and click Host
- Launch another machine and click Join
- You’re done!